Big Data Concepts Explained

1. Introduction

Big Data tools like Hive, Sqoop, Pig, and Spark are used to store, process, and analyze large volumes of data. These tools run on top of Hadoop or integrate with it.

2. Types of File Format

- CSV: Plain text, comma-separated
- JSON: Semi-structured, readable
- Avro: Row-based binary, supports schema
- Parquet: Columnar format, good for analytics
- ORC: Optimized columnar, efficient for Hive

3. Data Serialization

Serialization = Converting data to a storable/transmittable format.
Example in Python:

```
import json
data = {"name": "Ali", "age": 25}
json_string = json.dumps(data)
```

4. Importing MySQL and Creating Hive Table

Sqoop Command:

```
sqoop import --connect jdbc:mysql://localhost/employees --username root --password yourpass --table employee --hive-import --create-hive-table --hive-table default.employee_hive
```

5. Hive Table (Hivetb)

Hive Table Example:

```
CREATE EXTERNAL TABLE students (name STRING, age INT)
ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' LOCATION '/user/hive/warehouse/students';
```

6. Parquet with Sqoop

Sqoop can't directly import Parquet. Use Avro:

```
sqoop import --as-avrodatafile --target-dir /user/hive/employee_avro
```

7. Overview of Hive Query

Example:

```
SELECT name, age FROM employee_hive WHERE age > 25;
```

8. Getting Datasets for Pig

HDFS Upload:

```
hdfs dfs -put localfile.txt /user/hadoop/input/
```

Pig Script:

```
data = LOAD 'input.txt' USING PigStorage(',') AS (name, age);
DUMP data;
```

9. Basics of Apache Spark

PySpark Example:

```
from pyspark.sql import SparkSession
spark = SparkSession.builder.appName("Example").getOrCreate()
data = [("Ali", 25), ("Sara", 30)]
df = spark.createDataFrame(data, ["name", "age"])
df.show()
```

10. Spark Architecture

- Driver: runs main function
- Executors: run code
- DAG: task pipeline
- Action: triggers execution